

DC Hub Power Delivery Methodology

Audit-grade methodology specification for CBRE Research citation review.

Version 1.0 · 2026-06-04 · Jonathan Martone, DC Hub

SCOPE OF THIS DOCUMENT

- Months-to-energization (Power Delivery Index) — source data, refresh cadence, calculation, confidence intervals, known false-positive cases.
- AI vs traditional facility classification (derived_use field) — feature inputs, classifier output, confidence bucketing, published false-positive rates, acknowledged limitations.
- Shell-co M&A pattern matching — source corpus, pattern documentation, confidence thresholds, audit trail.
- Citation practice — recommended attribution string, methodology versioning, reproducibility guarantees, contact for compliance escalation.

OUT OF SCOPE FOR v1.0

International (non-US) queue · gas-to-grid heat-rate ranges · per-utility tariff lookups · ICE forward curves. Targeted for v1.1.

Months-to-energization, per market — calculation specification

How DC Hub produces the months-to-energization number for any tracked DCPI market.

SOURCE DATA

Live ingestion from the authoritative queue publishers

ISO/RTO interconnection queues (live):
PJM, ERCOT, MISO, SPP, CAISO, ISO-NE, NYISO
Vertically-integrated utilities (filing-based):
Duke (DEC, DEP), Southern (GA Power, AL Power, Mississippi Power), SRP, APS, TVA, BPA, AESO, Hydro-Quebec, NV Energy, PacifiCorp.

Per row: queue position, MW requested, queue entry date, current study phase (FS / SIS / FIS), COD target

CALCULATION

Median time-to-energization, study-velocity weighted

- For each market:
1. Filter to active large-load queue (≥ 20 MW)
 2. Compute median months between queue entry and last completed study transition (FS→SIS→FIS).
 3. Multiply by remaining-stage factor based on the 90-day rolling study-completion velocity in the parent ISO.

REFRESH CADENCE

Daily snapshot at 04:00 UTC

Endpoint: `/api/v1/interconnection-queue/snapshot`
Each row stamped `fetch_at` + `source_publish_date`.
Historical snapshots: 5 years retained, queryable.

Snapshot deltas surfaced via:
`/api/v1/interconnection-queue/diff?since=YYYY-MM-DD`

Source-document hash recorded per snapshot so any published number can be traced to the ISO's original

FALSE-POSITIVE HANDLING

Withdrawal decay + speculative filter

Speculative requests that withdraw inflate the P50.
Handled by withdrawal-decay weighting: each queue row gets a `continuity_weight = 1.0` at entry, decaying exponentially toward 0 if the request is withdrawn, reaches 0.0 at `withdrawal_date` + 180 days.

Markets with fewer than 8 completed reference projects in the trailing 24 months publish NULL with a `low_confidence` flag rather than a guessed value.

derived_use field — feature inputs, output, confidence bands

Closes the gap Gordon publicly named in H2 2025: "difficult to determine which facilities are designated for AI versus traditional storage or compute."

FEATURE INPUTS (heuristic stack)

For each facility, the classifier consumes:

- Density (kW/rack) — AI workloads start ≥ 30 kW.
- PUE published — traditional cooling-optimized ranges differ from liquid/immersion footprints typical of AI.
- Public press / SEC filings naming the use case (operator disclosures, tenant announcements).
- Construction permits + IT manufacturer disclosures (NVIDIA / AMD shipment correlations from public order data).
- Cooling type (air / liquid / immersion) where disclosed in air-permit filings or operator marketing.
- Power capacity uplift (≥ 50 MW announced expansion in 24 months → high prior for AI repurposing).

OUTPUT + CONFIDENCE BANDS

Output classes:

AI · Traditional · Mixed · Unknown

Each facility carries a confidence score 0.0–1.0:

high	≥ 0.85	published with strong claim
medium	0.60–0.85	published with caveat
low	< 0.60	collapsed to Unknown

Critical default: when confidence < 0.60 we publish Unknown rather than guess. Holdout-set false-positive rates (n=412 manually-labeled):

AI → claimed correctly	93.4%
Traditional → claimed correctly	91.1%
Unknown rate (refuse-to-guess)	8.2%

ACKNOWLEDGED LIMITATIONS (publish in any CBRE methodology footnote)

Greenfield announcements (unbuilt facilities) frequently lack disclosure on workload type. Default class is Unknown until construction permits or tenant disclosures land. Co-location operators serving multiple tenants are classified Mixed when any tenant disclosure indicates AI workload — under-counts traditional-only co-lo blocks. Refresh frequency: weekly re-classification on any new public signal.

Shell-co matching · recommended citation · reproducibility guarantees

How a number from DC Hub published in CBRE Research can be traced back to its source row in <10 seconds.

SHELL-CO M&A PATTERN MATCHING · /api/v1/transactions + /api/v1/utility-filings

Source corpus: 2,032 verified data-center M&A transactions (2018-present) + weekly utility-filing ingest across 22 grids.
Match logic: utility filing entity name + filing address ↔ deal corpus shell-co naming patterns + acquirer pipeline. Two confidence tiers:
explicit_match — named entity equality (highest confidence; surfaces immediately)
inferred_match — naming pattern + queue-position + capacity bracket (medium; flagged for review)

RECOMMENDED CITATION PRACTICE · CBRE RESEARCH METHODOLOGY FOOTNOTE

Use the same footnote convention CBRE uses for CoStar in office leasing reports:
DC Hub Power Delivery data layer (v1.0). /api/v1/interconnection-queue/snapshot. Refreshed daily.
Methodology: dchub.cloud/methodology/queue/v1.0 · Snapshot date: {YYYY-MM-DD}
Methodology version is tagged in every API response header (X-DC-Methodology-Version: v1.0) so a published number remains reproducible even after the methodology is updated.

AUDIT TRAIL + RUN-TIME REPRODUCIBILITY

Every published number traces to a specific row in interconnection_queue_snapshot + market_gas_pricing + transactions, stamped with fetched_at + source_publish_date + run_id.
Recovery procedure (compliance escalation): provide the published number + report date → DC Hub returns the source rows + ISO-published source URLs within 48 hours.
Compliance contact: Jonathan Martone · jonathan@dchub.cloud · 48-hour SLA on any data-traceability request from CBRE Research or CBRE Legal.